Project 2: The Librarian

CS496 AI Engineering, Fall 2026 · 4 Nov 2026 · **Live: http://158.158.120.189**

Ask a question about Tunisian law in Arabic or French. The Librarian finds the right article and answers from it, citing the law and article for every claim. If the laws don't contain the answer, it says "I don't know".

	result
Right article in the top 5	100% (main questions), 76% (paraphrased questions)
Answers correct / faithful to their sources	94% / 95% (after Week 8)
Unanswerable questions refused	22 of 24
Week 8	p95 latency −52%, for −2 points of correctness
Cost per question	$0.0002

1. The corpus and its users

- **Corpus:** laws from the *Journal Officiel* (iort.gov.tn), 2013–2026: **171 in Arabic, 179 in French**, about 12 million characters. Public, no personal data, approved by the instructor.
- **Left out:** loan and treaty ratifications (little content), and laws before 2013 (their PDFs extract as junk).
- **Users:** law students, lawyers, journalists and citizens looking for a rule, a deadline or a penalty.
- **Test questions:** 124 (`data/eval/qa.jsonl`): 50 facts asked in both languages, plus 24 questions the laws cannot answer. Each has a reference answer and its source article. A second set of **50 paraphrased questions** uses everyday wording, with no law numbers: the harder, more honest test.

2. How it works

`question → keyword search + meaning search → merge → reranker picks the best 5 articles → GPT answers with citations`

step	**choice**	**why**
Chunks	one per legal article (15,831)	a citation must name a law and an article
Index	flat files, exact search	16k chunks search in milliseconds; a vector database adds nothing at this size
Keyword search	BM25, Arabic and French normalised	questions often quote numbers and names
Meaning search	OpenAI embeddings (built in 3 min, $0.14)	finds the article even in the other language
Reranker	cross-encoder reads the top candidates	biggest accuracy gain, and the slowest step
Answer	`gpt-4o-mini` , only from the 5 articles, `[n]` after each claim	faithfulness over fluency

Details and the reasons behind each choice: `design.md` .

3. Retrieval results

Share of questions where the right article is found (main set / paraphrased set):

search	**in top 1**	**in top 5**
keywords only	87% / 12%	93% / 32%
meaning only	73% / 34%	92% / 66%
both combined	84% / 26%	94% / 58%
both + reranker	**97% / 52%**	**100% / 76%**

- **The reranker is the biggest gain:** it puts the right article first in 97% of cases.

- **Keyword search collapses on paraphrased questions** (32%), because users don't use the laws' words. Meaning search carries the system there.
- **Cutting by article beats fixed-size chunks** at the top ranks (97% vs 94% in top 1), and makes citations exact.

4. Answer quality

124 questions, graded by a second model (`gpt-4.1-mini`) against the reference answers.

	first version	**after prompt fix**
Correct	87%	**96%**
Faithful to the cited articles	not re-measured*	96%
Right article cited	99%	99%
Unanswerable questions refused	22 of 24	22 of 24
Answerable questions wrongly refused	0	0

* Our grader first checked answers against the wrong sources, which made faithfulness look like 85%. Once fixed, it was re-run on the improved version only.

- **Most errors were incomplete, not false.** The answer gave the fine but forgot it doubles for repeat offenders. Telling the model to include every condition, penalty and exception fixed most of them.
- **Some French questions were answered in Arabic.** A stricter language rule fixed it.
- **A "refuse when the search score is low" rule was dropped.** It looked perfect on the main set but would refuse 74% of the paraphrased questions.
- **Still failing:** 4–5 answers per 100, mostly budget figures given without their breakdown, and one petrol-price question answered from a fuel-tax table.

5. Week 8: faster, same accuracy

Goal: cut p95 latency or cost by 50%, losing at most 2 points of correctness.

The reranker took almost all the time, so it was the target. Each change was tested on the same 124 questions:

change	**p95 latency**	**cost**	**correct**	**kept?**
before (reranker reads 30 candidates)	—	$0.00023	96%	
reranker reads 10 candidates	**−52%**	same	**94% (−2)**	**yes**
reranker reads shorter text	−31%	same	93% (−3)	no
15 candidates + shorter text	−43%	−13%	95% (−1)	no: short of 50%
send 3 articles to GPT instead of 5	−7%	−28%	92% (−4)	no: answers less complete
cheaper model (`gpt-5-nano`)	—	same	empty answers	no

- **Result:** the latency target is met (−52%), and the accuracy limit is met exactly (−2). A clean re-run measured a loss of only 1 point. The cost target was not met: answers already cost $0.0002.
- **Caveat:** the laptop's speed changed during the runs, so the percentages are reliable but the seconds are not. Full table and method: `optimisation.md` .

6. Known failures and cost

Known failures:

- **Old laws and codes are missing.** Only laws from 2013 onward; no Penal Code, no Code of Obligations. Questions on drug penalties or contracts get "I don't know".
- **Amendments are not merged.** Each law is stored as published, so the system can cite an article that was later changed.
- **Tables.** Budget laws extract as loose numbers, so figures can come out without their breakdown.
- **Arabic and French versions sometimes differ** in article numbers, and some source files are mislabelled by the website.
- **Paraphrased questions:** the right article is in the top 5 only 76% of the time.

Cost (`results/cost_model.md`):

	cost
Build the index, once	$0.14
One question	$0.0002
100 questions a day (class demo)	about $0.60 a month + the VM
10,000 questions a day (100×)	about $60 a month in model fees; hosting grows with the reranker's CPU time, which Week 8 cut by half

Not done: feedback from real users. The site logs 👍 / 👎 , but no user round was run. All numbers come from our own 174 questions. What went wrong along the way: `postmortem.md` .